

Cleverscope CS548 calibration and accuracy

Summary

This instruction guides you through calibrating the Cleverscope CS548. We include a discussion on accuracy.

Introduction

You can either calibrate one channel at a time, or calibrate any number of channels together (1-4). The application will automatically work out which channels are being calibrated.

All CS548's include a switchable voltage reference (DC Cal) that can be used to calibrate individual channels. DC Ref is based on a $7.5V \pm 0.035\%$ voltage reference, and switches between 0, 0.6818 and 7.5V.

Calibrating one Channel at a time using DC Cal (Cal Ref v1.0 units)



Power up the CS548, plug in the USB, and get the unit capturing with the Cleverscope4 application. Allow the unit to capture for at least 10 minutes, to warm up. The internal temperature is stabilized at 40 deg C.

Using a single scope probe, set to 1x, use Settings/Calibrate Cleverscope, and choose the Single Channel tab.

Find the DC Cal and 0V terminals on the right hand side of the CS548 front panel.

Either use the Probe as shown (min visible voltage noise) or use the Probe clip. It does not matter which.

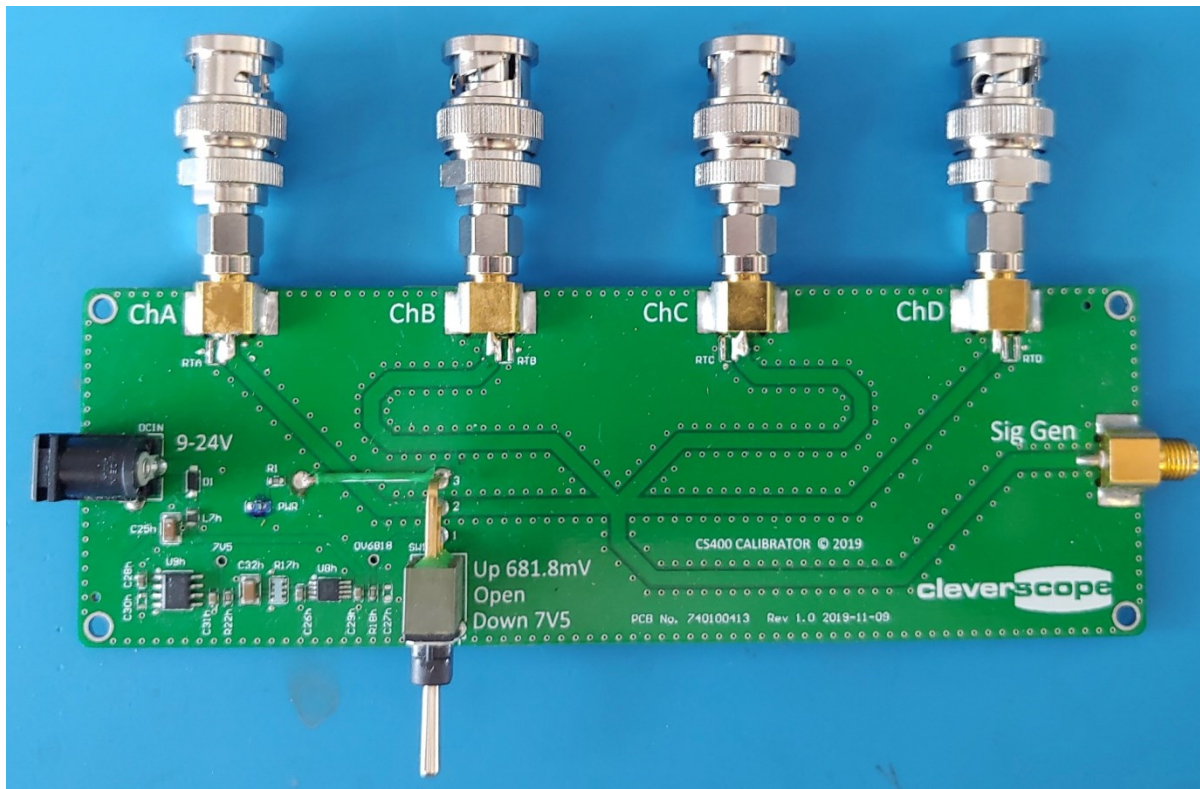
Connect the Probe to Channel A. Click on 'Calibrate'. After a short time the Cal Complete will turn on green.

Next complete this process for Channels B - D. Click Finish at the end.

If there is an error, the Cal Error led will light red. Makes sure both the probe tip and ground clip are connected. Try again.

Calibration of 1-4 channels using the Cleverscope Calibrator

You can use the Cleverscope Calibrator to calibrate all four channels of the CS548 together. You will need to obtain a Calibrator from Cleverscope.

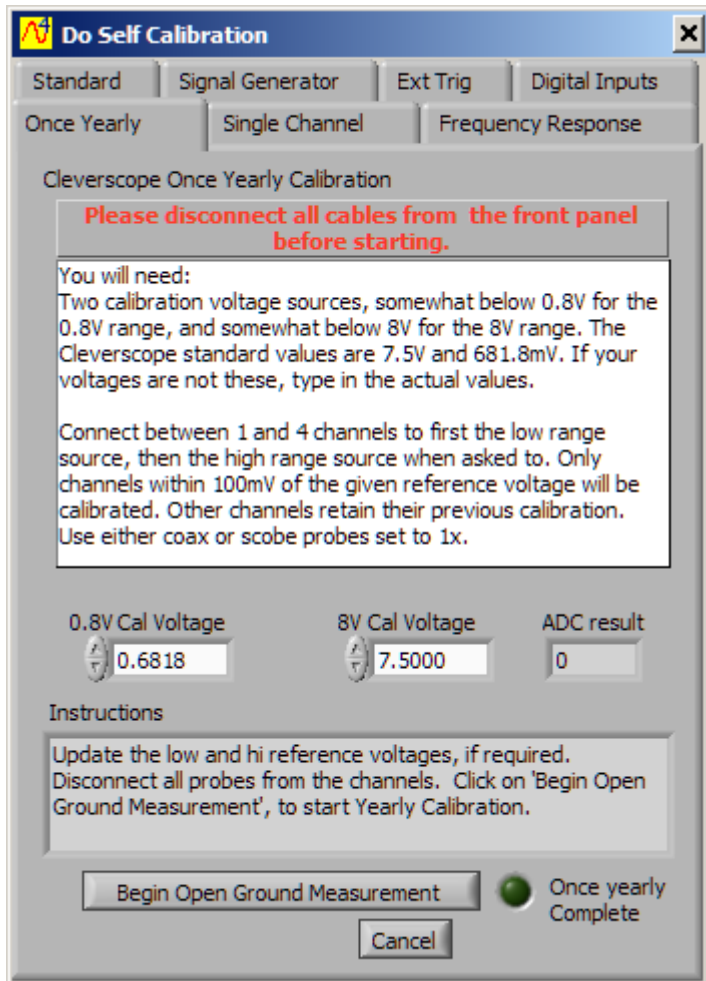


The calibrator has a 9-24V DC input to power the reference. You can use a Cleverscope 9, 12 or 19V power supply for this (centre pin positive). The three way switch chooses between Open (center position), 681.8mV (switch up) or 7.5V (switch down). There is also a sig gen connection for calibrating the signal generator, or for checking channel time alignment. The calibrator uses the same $7.5V \pm 0.035\%$ voltage reference as the CS548.

Plug the Calibrator into the CS548, and attach a power supply to it. Allow it to warm for 10 minutes:



Run the scope at least 10 minutes to warm up. Choose Settings/Calibrate Cleverscope, and choose the Once Yearly Tab:



Carry out these steps:

1. Have the switch in the middle, open position, and click the 'Begin Open Ground Measurement' Button.
2. Once the Button name changes to 'Begin 0.8V Range Cal Measurement'; flip the switch up to the 0.6818V position. Click the Button.
3. Once the Button name changes to 'Begin 8V Range Cal Measurement', change the switch down to the 7.5V position. Click the Button. On completion the 'Once yearly Complete' led will light. Calibration is complete.
4. Click Finish to exit. Remove the Calibrator. The CS548 is ready to use.

Calibration without the Cleverscope Calibrator.

Follow the same sequence as using the Cleverscope Calibrator, but use an external voltage reference instead – you will need one about 0.68V and another about 7.5V. Enter the exact voltages into the Do Self Calibration panel.

Follow the same sequence, but apply open, 0.6818 and 7.5V to the input, or inputs as you click the Button to change to the next voltage.

Calibrating the Signal Generator

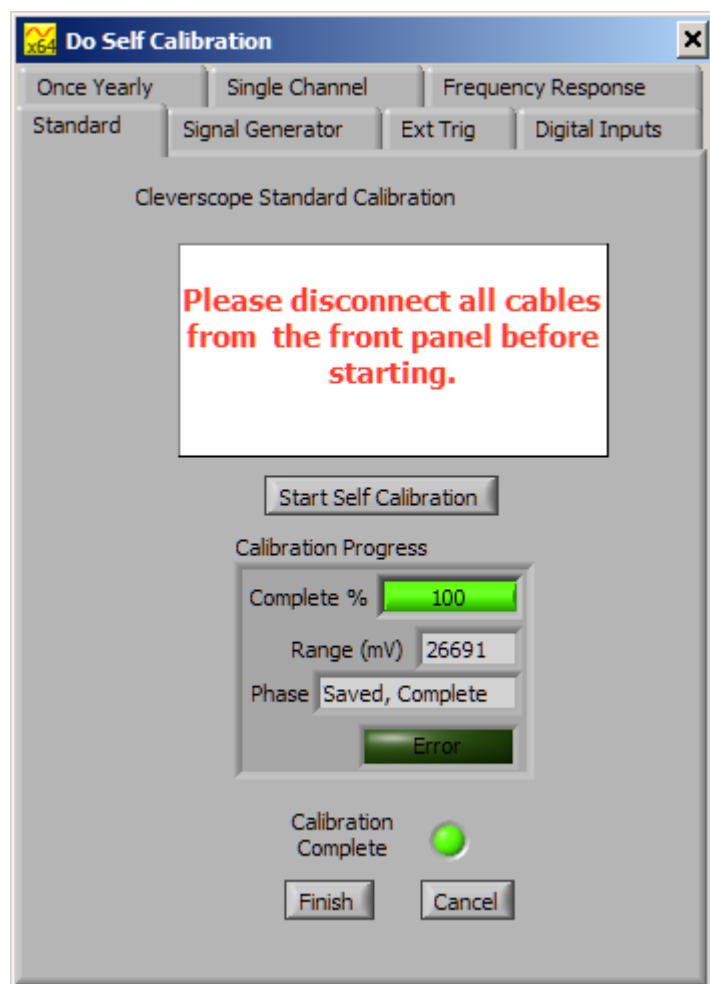
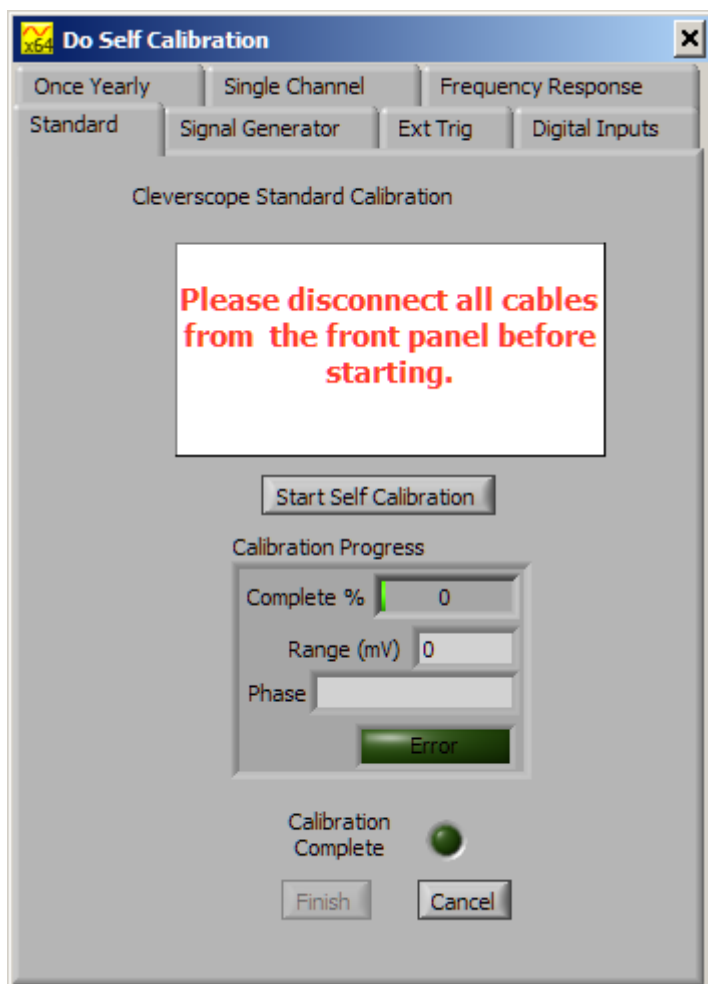
To Calibrate the signal generator, connector a 1x probe with a BNC adaptor from the accessories plastic bag, or Coax from the signal generator output (usually the filtered output) to Channel A. Click the Signal Generator tab on the Do Self Calibration panel. Follow the instructions. You can also connect the Signal Generator to the right hand socket on the Cleverscope Calibrator:

If using the 4 way calibrator, make sure you leave the switch in the centre position. It does not matter if the Calibrator is powered.



CS548 Self Calibration

The CS548 includes a voltage reference inside each channel. During Single Channel calibration, or Once Yearly calibration, the internal reference is referred to the external reference, and gain and offset for each reference is calculated. The internal reference is used during 'Standard' Calibration to recalibrate the channel, taking into account any offsets developed by the signal chain. To carry out Standard Calibration, remove all probes from the CS548, allow the unit to warm up for 10 minutes, and then run Settings/Calibrate Cleverscope/Standard, and click on 'Start Self Calibration'. While running you will see each phase of the calibration being carried out, with the Calibration Complete led lit when the all is complete. Click Finish to exit. It does not take long.



The CS548 uses active fan speed control to maintain the Digitizers at 40 deg C. This ensures that the internal calibration reference is always operating at the same temperature. The long term stability (1000 Hours) of the LM3040-2.5 used is 40 ppm, and we have found that the CS548 still self calibrates to better than the $\pm 0.15\%$ after a year.

CS548 Accuracy

The CS548 includes built in voltage and time references, brought out for external use. These are:

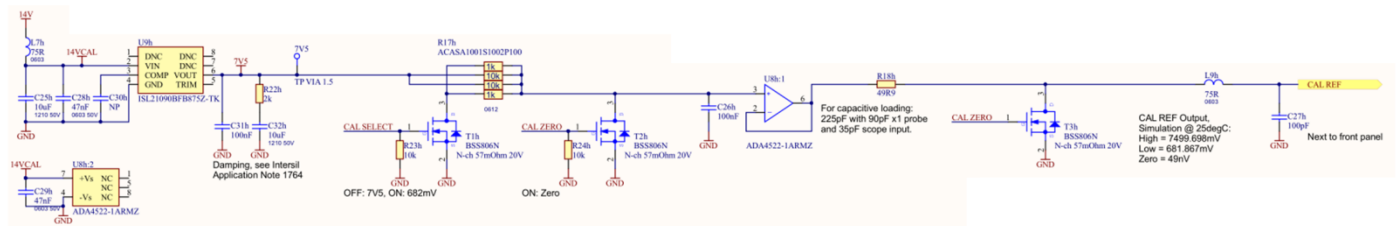
DC Cal

This output uses a buffered Intersil ISL21090BFB875Z-TK 7.5V reference. A precision ratiometric resistor array is used for the 0.6818V reference, and a shorting FET for the 0V output. The output is buffered by an Analog Devices ADA4522 precision op-amp (typ 0.7uV, max 5uV offset) with low drift (2.5nV/deg C).

Specifications are:

- Output voltages: 0, 0.6818 and 7.500 V.
- Accuracy: $\pm 0.035\%$ (± 2.6 mV) (max), $\pm 0.025\%$ (± 2 mV) (typ).
- Temperature drift: 52uV/deg C (max).
- Noise: 6.2uVp-p (0-10Hz), and 4.8 uV_{RMS} (broadband).
- Stability < 20 ppm (1000 Hrs).

The schematic is shown below (expand the PDF to see the detail):



Probe Comp

The probe comp output BNC is a programmable FPGA based pulse generator. It uses a 233.33333 MHz clock to generate output patterns. The 233.33333 MHz is PLL derived from an NDK NT2016SA-26.000000MHz-NBG2 clock oscillator. A Skyworks Si5386a low phase noise, jitter cleaning PLL is used. The same Si538A is used for all ADC clocks, and the FPGA clock.

Specifications are:

- Accuracy: ± 2 ppm max, ± 1 ppm typ
- Temperature drift: ± 0.5 ppm max over -30 .. +85 deg C temperature range
- Long term stability: ± 1.0 ppm max over 1 year
- Short term stability: 1.0 ppb ($\tau = 0.1$ sec)
- Phase Noise: 118 dBc/Hz @ 100Hz offset
- Jitter: 72 fs typ (12 kHz–20 MHz)

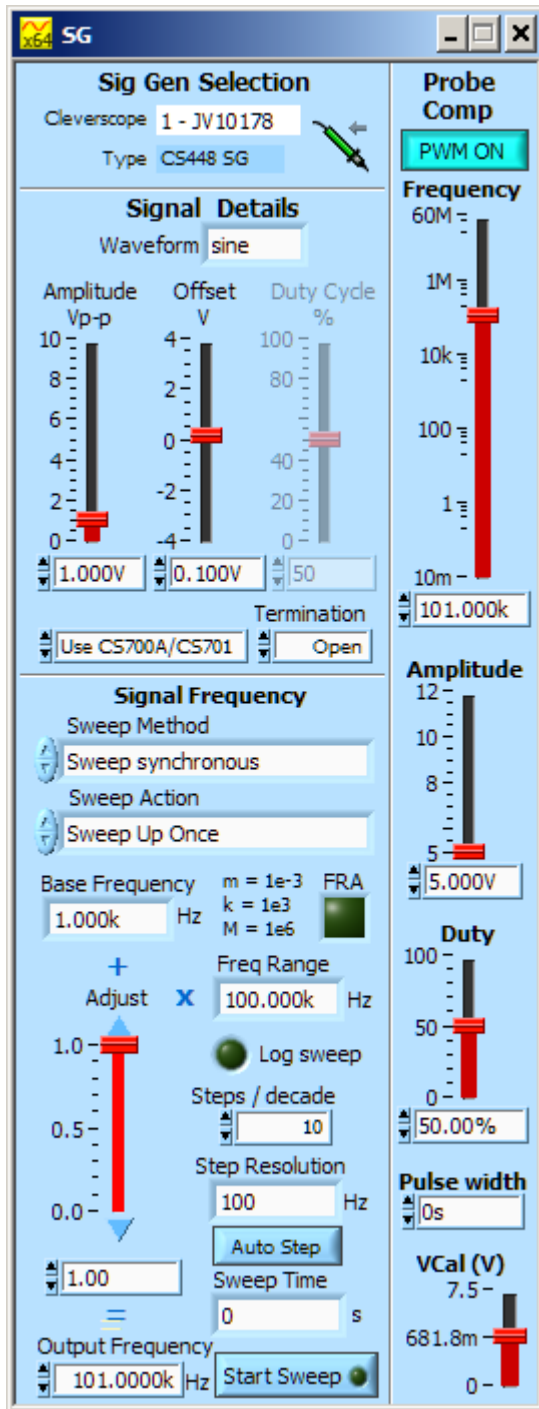
Voltage Measurements

Cleverscope uses a calibrated HP34401A 6.5 digit system voltmeter to measure the DC Cal voltages of every CS548 made. The HP34401A specification is:

- Accuracy: $\pm 0.0035\%$ (reading) + $\pm 0.0005\%$ (range) max over 1 year.
7.5V measurement: $260\mu\text{V} + 50\mu\text{V} = \pm 310\mu\text{V}$
0.6818 V measurement: $21.7\mu\text{V} + 5\mu\text{V} = \pm 26.7\mu\text{V}$
- Temperature drift: $\pm 0.0005\%$ / deg C

Settings the CS548 DC Cal Output Voltage

Use View/Sig Gen Controls to display the Sig Gen Controls:



Expand the panel to see the Probe Comp and Vcal controls by clicking the  button.

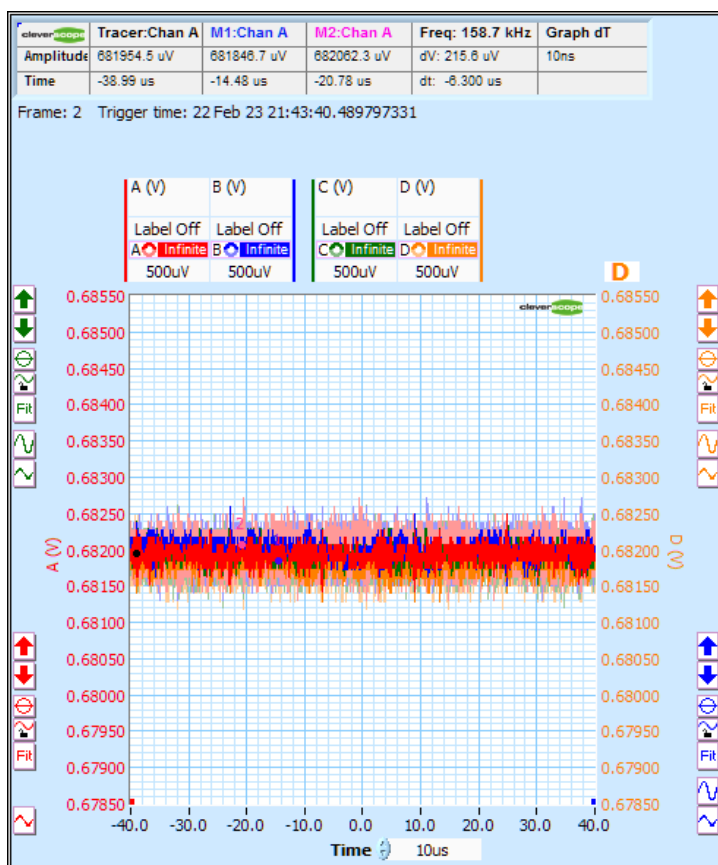
Set the DC Cal voltage (Vcal) using the control. You can set it to 0, 681.8mV and 7.5V.

CS548 Display Accuracy

Use the CS548 to measure DC Cal outputs to estimate display accuracy, and variation over time. The DC Cal output can be used as the source, or a CS548 Calibrator be used instead. Our measurements:

- 24 Hour variation: ± 1.0 mV max on 8V range (± 0.013 %)
 ± 100 μ V max on 0.8V range (± 0.013 %)
- 1 year variation ± 12 mV max on 8V range (± 0.15 %) after Self Calibration
 ± 1.2 mV max on 0.8V range (± 0.15 %) after Self Calibration

The CS548 application can be used to log and display the variation over time. Here the scope display has infinite persistence turned on, with results after 24 hours (a 10 MHz filter was used - Settings/Acquisition Settings, to avoid local radio station noise):



The information display (View/Signal Information) displays statistics:

Signal Information				
Signal Information				
Function	Chan A	Chan B	Chan C	Chan D
DC	681.9 mV	682.0 mV	681.9 mV	681.8 mV
RMS	681.9 mV	682.0 mV	681.9 mV	681.8 mV
Max	682.5 mV	682.6 mV	682.5 mV	682.3 mV
Min	681.3 mV	681.5 mV	681.4 mV	681.3 mV
Pk-Pk	1.186 mV	1.085 mV	1.080 mV	967.0 uV
Std Dev	102.6 uV	108.1 uV	108.3 uV	111.6 uV
Period	60.00 ns	55.00 ns	125.0 ns	237.5 ns
Fundamenta Frequency	17.54 MHz	63.27 kHz	17.54 MHz	16.46 kHz
Fundamenta Peak amp	24.16 uV	31.22 uV	22.35 uV	45.62 uV
Pulse Length	234.1 ns	47.80 ns	103.0 ns	56.86 ns
Duty Cycle	50.01 %	36.36 %	60.00 %	83.16 %

Here we see a dispersion of about $\pm 0.8\text{mV}$ on the Scope Graph, including noise.

DC reference measurement examples

Using the HP34401A to measure the three output voltages, 0, 681.8mV and 7.5V with an arbitrary CS548.

0V output from DC Cal:



Output = 64 uV.

681.8mV output from DC Cal:



Output =
681.864mV

Error = +0.009%

7.5V output from DC Cal:



Output =
7.49971V

Error = -0.004%

Conclusion

The CS548 includes the voltage and time references to achieve accuracy much better than a standard oscilloscope. It includes the self calibration resources to quickly achieve specified accuracy. The application includes the measurement capability to allow the user to verify the accuracy is as specified.